Bucket Brigade: A Parametric Cooperative-Competitive Benchmark with Computable Nash Equilibrium Structure

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Abstract

Canonical cooperative-MARL benchmarks (Overcooked, Hanabi, SMAC, Melting Pot, MAgent, PettingZoo) trade equilibrium transparency for environment richness: no Nash equilibrium is computable for them, so when a new algorithm beats PPO the reader cannot tell whether it converged to the "right" joint policy or merely to a Pareto-different one. *Bucket Brigade* inverts this trade: a finite parametric cooperative-competitive stochastic game (2304 states and 8 actions per agent at the minimal parameterization) whose three scalars (β, κ, c) control equilibrium structure continuously. We derive closed-form boundary inequalities for three NE regimes via a single-house mean-field reduction and test them against a 39-cell empirical phase diagram: the qualitative phase order (collapse → symmetric → mixed → asymmetric) is correct, the quantitative κ-thresholds are off by 3–10× (a gap we own and attribute to three named biases), and a fourth `mixed` class emerges that the bound does not predict. A 37-cell PPO trainability sweep recovers the predicted class ordering under a per-cell NE-anchored metric (the previously-used single-cell baseline inverts it—a standalone methodological lesson). We retire two intuitive trainability predictors in sequence: conditional entropy of the NE profile does not predict per-cell gap closure (a β-invariance argument rules out *every* NE-profile statistic, and a 20-seed noise buy-down re-verifies the retirement), and the replacement coordination-threshold (k^*) hypothesis is falsified in its binary form; the surviving $k^*{=}k_{\max}$ failure-zone observation is post hoc and κ-confounded, and is flagged as such. On the social-trap scenario `rest_trap-v1`, whose equilibrium characterization is closed, a scripted specialist team demonstrates a machine-verdicted ≈80/step learnability gap: standard-budget PPO is statistically indistinguishable from random play, and a 16×-budget ladder yields exactly one marginal, variance-driven escape with no mean improvement. We position Bucket Brigade as a *methodological complement* to the surveyed benchmarks: to our knowledge the only published *parametric* MARL family in which "did the algorithm converge to the right equilibrium?" admits a ground-truth answer across a controlled grid.

1 Introduction

Multi-agent reinforcement learning evaluates new algorithms against benchmarks whose Nash-equilibrium structure is not characterizable in closed form. Overcooked-AI (Carroll et al., 2019), DeepMind's Melting Pot (Leibo et al., 2021), the Hanabi Challenge (Bard et al., 2020), SMAC and SMACv2 (Ellis et al., 2023; Samvelyan et al., 2019), MAgent (Zheng et al., 2018), and the PettingZoo MPE suite (Terry et al., 2021) between them cover small cooperative coordination, partial observability, scripted-opponent micromanagement, and population-scale competition. None of them ship with a method that produces the equilibrium strategy profile of the as-designed environment. When PPO fails to learn a coordinated policy on Overcooked, the community is left to debate whether the failure is the algorithm or the absence of a clean attractor for self-play; neither hypothesis can be rejected because no oracle exists for "what should the learned policy look like."

We introduce *Bucket Brigade*, a finite parametric cooperative-competitive stochastic game designed to invert this trade. At its minimal parameterization the state space is 2304 states and the per-agent action space is 8, so the one-shot stage game is enumerable and standard equilibrium-computation tools actually run. Across a smooth three-scalar parameter family (β, κ, c) the equilibrium structure transitions from symmetric all-Work to asymmetric 1-Worker (the textbook free-rider problem) to no-pure-NE collapse, with a fourth empirically-observed regime – mixed, where both symmetric and asymmetric pure NE coexist – appearing at high κ on the higher-c rows of the grid. The ground-truth target for each parameter cell is known, so a failed PPO run is attributable to the algorithm rather than to ambiguity about what convergence would mean. The pitch is exactly a *methodological complement* to the six canonical benchmarks above: those exist for studying open-ended cooperation, generalization, and scale; Bucket Brigade exists for studying equilibrium-convergence quality on a controlled grid.[1]

Contributions. (1) A self-contained formal specification of the Bucket Brigade game (§2), recovering Volunteer's Dilemma, the N-player Public Goods game, and Stag Hunt as named limits. (2) Closed-form boundary inequalities (§3) for the three NE regimes via a single-house mean-field reduction, together with an explicit accounting of where the closed-form thresholds disagree with empirics and where a fourth (mixed) class emerges that the bound does not predict. (3a) A protocol and the first empirical results (§4) testing PPO trainability against the NE phase diagram on the full 37-cell grid: the PPO ordering recovered by a per-cell NE-anchored metric is consistent with the analytical prediction symmetric > mixed > asymmetric > collapse, and a separate 4×-budget sweep on the no-convergence cells worsens, not improves, gap-closure – PPO's failure on those cells is structural, not budget-limited. (3b) The per-cell NE-anchored metric itself, surfaced as a standalone methodological contribution: on this paper's data, the previously-used single-cell baseline inverts the per-class ordering (§4), so per-cell calibration is not a cosmetic refinement but a necessary correction for cross-cell PPO comparison whenever the random-policy return varies across cells. (3c) A double negative result on per-cell trainability predictors (§4): we retire two intuitive predictors in sequence and identify the honest remaining signal. First, conditional entropy of the converged NE profile is uncorrelated with per-cell gap closure, and a β-invariance argument rules out *every* statistic of the NE profile as an explanation of within-(κ, c)-column variance (a 20-seed noise buy-down on an 8-cell subset shrinks CI half-widths 2.0–3.6×, flips none of the 8 trainability verdicts, and re-verifies the retirement). Second, the replacement coordination-threshold hypothesis—PPO trains where $k^*{=}1$ and fails where coordinated multi-agent deviation is required—is falsified in its binary form by the full-grid join against a downstream per-cell k^* oracle: $k^*{=}1$ and $k^*{\geq}2$ columns are indistinguishable on the pooled NE gap (rank-biserial 0.00, exact Mann–Whitney $p{=}0.54$). The remaining signal is exploratory and we flag it as such at every mention: the $k^*{=}k_{\max}$ columns coincide exactly with the untrainable zone, but the split is post hoc, and k^* is a pure function of κ on this grid, so its content is observationally equivalent to a κ effect. (4) A measured social-trap hardness result (§5): on `rest_trap-v1` a fully-scripted specialist team scores 386.60/step against a 302.87/step uniform-random baseline, while 20-seed PPO at the standard budget is statistically

[1] Page budget: this is the extended version of record; the target venue's 4-page workshop body is a strict subset of it. The 4-page version keeps §1, the §2 environment contract, the §3 boundary inequalities with Figure 1, the §4 class-ordering and retired-predictor results with Figure 2, the §5 anchor ladder (Table 2), and the §6 positioning, each in compressed form; the formal specification (Appendix A), the derivation and bias accounting (Appendix B), Tables 1 and 3, and the detailed predictor post-mortems of §4 move to the supplementary appendix, outside the 4-page body count.

indistinguishable from random and a 16×-budget run separates from random only marginally and without mean improvement – a machine-verdicted ≈80/step learnability gap, grounded in a measured anchor ladder and a pre-registered verdict rule rather than an ad-hoc fraction gate. The scenario's equilibrium characterization is closed: the symmetric double oracle cycles across two independent runs (and the seeded run's final mixture is measurably far from equilibrium), so the asymmetric equilibrium stands. (5) An honest positioning (§6, §7) against the canonical benchmarks: Bucket Brigade is small and artificial by design, and is not a substitute for any of them. Throughout, we treat the analytical NE characterization as ground truth and the PPO trainability sweep as a falsification test, in that order.

2 The Bucket Brigade environment

Players and topology. A finite set of agents $\mathcal{I}=\{1,\dots,N\}$ inhabits a cycle graph $\mathcal{H}=\mathbb{Z}/H\mathbb{Z}$ of H houses with neighbour edges. Each house has status $h_h \in \{\textsc{Safe}, \textsc{Burning}, \textsc{Ruined}\}$; Ruined is absorbing. The default "10-house" scenario fixes $H=10, N=4, T_{\min}=12$; the minimal scenario fixes $H=2, N=4$.

Action and observation. Each agent i chooses $a_{t,i}=(a^{\text{house}}, a^{\text{mode}}, a^{\text{sig}}) \in \mathcal{H} \times \{0,1\} \times \{0,1\}$: a target house, a Work/Rest bit, and a one-bit broadcast signal observable to all agents on the next night. Per-agent action cardinality is $4H$ (8 at $H=2$; 40 at $H=10$). Joint-action cardinality at the minimal parameterization is $8^4=4096$. Observation is symmetric perfect monitoring with one-step signal delay: every agent observes the full state plus the previous joint $(\ell_{t-1}, \alpha_{t-1}, \zeta_{t-1})$. There is no partial observability of houses or agent positions; the signal channel is non-binding cheap talk.

Transition dynamics. A single night consists of seven sequential phases applied in a fixed, load-bearing order: (1) signal write, (2) location/mode write, (3) extinguish, (4) burn-out, (5) spread, (6) spontaneous ignition, (7) reward and termination check. In the extinguish phase, if $w_h(a)$ agents Work at a Burning house, the house transitions to Safe with probability $1-(1-\kappa)^{w_h(a)}$ (independent-workers model). Every Burning house that survives the extinguish phase transitions deterministically to Ruined in the burn-out phase. The spread phase then ignites Safe neighbours of Burning houses with probability β per edge; under the canonical order, the burn-out phase has already cleared all Burning sources, so spread is inert in the within-night dynamics and contagion enters only via the spontaneous-ignition phase 6 at rate ρ. (A contagion-active variant that swaps phases 4 and 5 is available; the phase-diagram analysis below uses the canonical order.)

Reward structure. The per-agent reward decomposes into three independently-parameterised terms: a private work/rest cost ($-c$ for Work, $+c_{\text{rest}}=0.5$ for Rest; the free-rider gradient is the $c+c_{\text{rest}}$ asymmetry), a per-house ownership term (r_{own} paid on save, p_{own} recurring on every Ruined night), and a team welfare term proportional to the fraction of Safe and Ruined houses. Ownership is assigned round-robin (owner$(h)=h \bmod N$). The asymmetric reward variants in the calibrated family let $r_{\text{own}}, r_{\text{other}}, p_{\text{own}}, p_{\text{other}}$ be set per-agent, breaking exchangeability.

Relationship to canonical templates. Bucket Brigade recovers, as named limits or restrictions, several textbook templates: the N-player Volunteer's Dilemma (Diekmann, 1985) at $H=1$; the N-player Public Goods game (Ledyard, 1995) when ownership weights collapse to uniform; Stag Hunt (Skyrms, 2003) in the $\kappa\to 1$ limit; and a free-rider problem at the parameter cells where the only NE is asymmetric (the "rest-trap"). Formal proofs and a complete notation table appear in Appendix A; §2 above is the minimal contract a reader needs to follow §3.

3 Equilibrium structure: closed-form boundaries

The central contribution of this paper is a closed-form characterisation of which Nash-equilibrium structure appears as a function of (β, κ, c) in the 4-agent game. We derive it via a single-house mean-field reduction that collapses the H=10 ring to a representative single house with all N=4 agents present, absorbs multi-step contagion and ignition dynamics into a per-night survival coefficient, and reads off the NE candidates as algebraic inequalities.

Reduction. Fix the `minimal_specialization` reward tuple

$$W = (R_{\text{team}}, P_{\text{team}}, r_{\text{own}}, r_{\text{other}}, p_{\text{own}}, p_{\text{other}}) = (10, 10, 50, 0, 100, 0),$$

with ρ=0.02 and T_{min}=12. With k workers at the house on a given night, the per-night ruin probability for an initially-SAFE house is $q(k)$=$\rho(1-\kappa)^k$ (under the canonical phase order, where β does not enter the leading-order single-night payoff). Integrating the lost per-house reward stream over T_{min}=12 nights and collecting terms yields

$$\mathbb{E}[\text{per-house reward loss} \mid k] = A \cdot \rho \cdot (1-\kappa)^k,$$

with A=$r_{\text{own}}T_{\text{min}} + p_{\text{own}}T_{\text{min}} + P_{\text{team}}T_{\text{min}}/H$=1812 and survival coefficient $\tilde{A}$:=$A\rho$=36.24. The marginal benefit of adding the k-th worker is $\Delta(k)$=$\tilde{A} \cdot \kappa \cdot (1-\kappa)^k$.

The three NE candidates. Writing c_{gap}:=$c + c_{\text{rest}}$ (so c_{gap}=1.0 when the driver sets c=0.5), the deviation comparisons against the per-night stage game under stationary opponent policies yield three inequalities.

The *symmetric all-Work* NE exists iff a unilateral deviation from k=4 to k_{-i}=3 does not pay:

$$\tilde{A} \cdot \kappa \cdot (1-\kappa)^3 \;\geq\; c_{\text{gap}}. \tag{S}$$

The *asymmetric 1-Worker* NE exists iff the lone worker prefers WORK (k_{-i}=0 deviation) and each free-rider prefers REST (k_{-i}=1 deviation):

$$\tilde{A} \cdot \kappa \;\geq\; c_{\text{gap}} \quad \text{AND} \quad \tilde{A} \cdot \kappa \cdot (1-\kappa) \;<\; c_{\text{gap}}. \tag{A}$$

The *no-pure-NE collapse* regime holds whenever even a lone worker cannot recover their cost, so neither (S) nor (A) can be satisfied:

$$\tilde{A} \cdot \kappa \;<\; c_{\text{gap}}. \tag{C}$$

Predicted thresholds. Substituting $\tilde{A}$=36.24, c_{gap}=1.0 into (S)–(C): the collapse boundary (C) sits at $\kappa \approx 0.028$; the symmetric NE exists on $\kappa \in [0.030, 0.65]$ via (S); the asymmetric NE exists for $\kappa \gtrsim 0.972$ via (A). The qualitative phase order κ-small→κ-large is collapse → symmetric_only → mixed → asymmetric_only, and the derivation is structurally β-independent under the canonical phase order.

Predicted vs. observed thresholds. The empirical phase diagram, computed by a heterogeneous Double-Oracle solver over continuous heuristic-parameter strategies on a 39-cell grid ($\beta \in \{0.1, 0.5, 0.9\} \times \kappa \in \{0.1, 0.3, 0.5, 0.7, 0.9\} \times c \in \{0.5, 1.0, 2.0\}$, less the six unsampled c=0.5 cells at $\kappa \in \{0.3, 0.7\}$—the two mid-κ columns of that panel, across all three β; Figure 1), recovers the qualitative phase order and approximate β-independence at verdict level: 12 of the 13 (κ, c)

rows with ≥ 2 β samples show an identical verdict across all sampled β. The single splitting row is (κ=0.9, c=0.5), where β=0.1 yields mixed while $\beta \in \{0.5, 0.9\}$ yield asymmetric_only.

An important caveat, established after the sweep by mechanistic inspection of the engine (repo issue #442): in the bernoulli extinguish mode the sweep uses, β-invariance is not merely a leading-order prediction of the reduction – it is *exact by construction.* Under the canonical phase order the burn-out phase clears every BURNING house before the spread phase runs, so spread is a guaranteed no-op and, because it draws zero RNG in this mode, the dynamics *and the RNG stream* are bit-identical across β. Verdict-level β-invariance therefore cannot confirm the reduction; it is a consistency check on the solver. The useful reading is the converse: any residual cross-β difference at fixed (κ, c) is a direct measurement of double-oracle solver nondeterminism. The lone splitting row above is exactly such a measurement – not an environment anomaly – and the same probe shows the solver's *payoffs* carry noise as well: at (κ=0.9, c=0.5) the β=0.1 run returns equilibrium payoff 80.9 against 72.0 at $\beta \in \{0.5, 0.9\}$ for the bit-identical environment (`experiments/nash/phase_diagram/results.json`). Equilibrium *payoff* values from the solver should accordingly be read with an unquantified solver-noise error bar; verdict classes, which agree on 12/13 rows, are more robust. (Whether the solver's converged asymmetric profiles are themselves ε-NE under common-random-number re-evaluation is an open question, tracked as repo issue #459.) The grid surfaces *four* empirical classes: in addition to symmetric_only (n=12), asymmetric_only (n=11), and no_convergence (n=6), a mixed class (n=10) appears in which the solver returns both symmetric all-Work and asymmetric 1-Worker profiles as un-deviable, primarily at κ=0.9 on the $c \geq 1$ rows and at κ=0.5, c=2.0. The closed-form bound (S)–(C) does not predict this fourth class at the higher-c regime where it emerges: the derivation pivots on $\tilde{A}\kappa(1-\kappa)^k$ being above or below c_{gap}, which excludes the simultaneous-existence pattern. Tightening the reduction to predict the mixed boundary in closed form is in scope for future work.

Table 1 summarises the per-κ predicted vs. empirical comparison on the full 39-cell grid. The qualitative agreement is strongest at $\kappa \in \{0.3, 0.5, 0.9\}$ (predicted class is the modal empirical class) and weakest at $\kappa \in \{0.1, 0.7\}$ (predicted class fails to appear at all in the empirical row): the same collapse-threshold and asymmetric-threshold disagreements the prose above identifies.

κ	Predicted class	Empirical distribution	Predicted-class share
0.1	symmetric_only	6 collapse, 3 asymmetric	0/9
0.3	symmetric_only	6 symmetric	6/6
0.5	symmetric_only	6 symmetric, 3 mixed	6/9
0.7	mixed/transition	6 asymmetric	0/6
0.9	mixed/transition	7 mixed, 2 asymmetric	7/9

Table 1: Predicted vs. empirical NE class on the full 39-cell grid, broken down by κ. "Predicted class" is the band identified by (S)–(C) at the centre of each κ value, taking the bordered case as the named single class for readability. "Empirical distribution" counts the verdicts of the heterogeneous Double-Oracle solver across all (β, c) cells at that κ; the row totals are 9 except at $\kappa \in \{0.3, 0.7\}$ (6 cells each: the c=0.5 panel was not sampled at those two κ columns). The reduction predicts the modal empirical class on 3/5 κ rows. The two off-rows ($\kappa \in \{0.1, 0.7\}$) are the collapse-threshold and asymmetric-threshold disagreements documented in the prose above.

The *quantitative* κ-thresholds also disagree: the empirical collapse boundary sits near $\kappa \approx 0.1$–0.3 (3–10× above the predicted 0.028), and the empirical asymmetric-NE onset sits near $\kappa \approx 0.7$ (below the predicted 0.972). The derivation gets the phase order right on the 39-cell grid but the thresholds wrong by 3–10×, the same gap we owned at the 7-cell preview.

We *own* this gap. The mean-field reduction has three identifiable systematic biases. (i) *Ring*

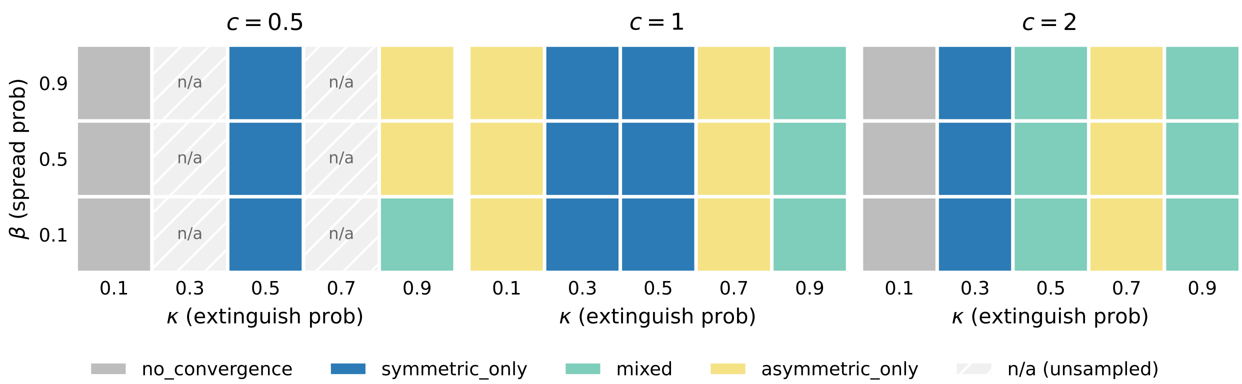

Figure 1: Empirical NE phase diagram for 4-agent Bucket Brigade across (β, κ, c) at ρ=0.02, $T_{\min}$=12 under the `minimal_specialization` reward tuple. Three panels, one per cost $c \in \{0.5, 1.0, 2.0\}$; within each panel rows are β and columns are κ. Each cell is the verdict of a heterogeneous Double-Oracle solver over the continuous heuristic-parameter strategy space (Hero, Firefighter, FreeRider, ...); the six unsampled cells (the $\kappa \in \{0.3, 0.7\}$ columns of the c=0.5 panel) are hatched. Four regimes appear across the grid: no_convergence at low κ (collapse), symmetric_only at moderate κ (Hero NE dominates), asymmetric_only at high $\kappa, c \lesssim 1$ (1 Hero + 3 Firefighters; the rest-trap), and a fourth mixed regime (both pure NE coexist) primarily at κ=0.9, $c \geq 1$. The grid is approximately β-independent at the verdict level (12 of 13 (κ, c) rows identical across β; the lone splitting row is (κ=0.9, c=0.5)). Note that in the bernoulli extinguish mode used here the environment is bit-identical across β (repo issue #442), so the β axis functions as a solver-noise probe rather than a test of (S)–(C); the κ thresholds remain substantially different from where the closed-form derivation places them.

locality: the derivation puts all four agents at one house, but empirically four agents distribute across ten houses, so the effective per-house defence is $1-(1-\kappa)^{0.4}$ rather than $1-(1-\kappa)^4$—roughly a 4× weaker defence at κ=0.5, which shifts the collapse boundary substantially upward into the empirically-observed range $\kappa \in [0.1, 0.3]$. (ii) *Per-agent ownership rewards*: the analytical A coefficient lumps r_{own} and p_{own} into the survival term, but the per-agent ownership reward only fires on the agent's own house, scaling the off-own-house contribution down by a factor of roughly $1/H$. (iii) *Heuristic strategy space*: the empirical solver's "free-riders" are `Firefighter` agents at `work_tendency`=0.9, not literal Rest agents, so the effective k_{-i} in the lone-worker's deviation comparison is closer to 3.7 than to 1. None of these corrections are derivable in closed form without losing the algebraic tractability that motivated the reduction; the natural extensions are named in §7.

Headline finding. The mean-field reduction correctly predicts *two* structural properties of the empirical phase diagram—the qualitative phase order and the existence of a no-pure-NE collapse regime at small κ—neither of which would be derivable without the closed-form derivation. (A third prediction, β-independence under the canonical phase order, is correct but turns out to be true by construction in the sweep's bernoulli extinguish mode, per the caveat above, so we no longer count it as an empirical confirmation.) The quantitative κ-thresholds are off by 3–10×, and a fourth empirical class (mixed) appears at the higher-c rows that the closed-form bound does not predict; we report both as the paper's claim, not as weaknesses to hide. The qualitative phase order furthermore transfers to the PPO trainability sweep under a per-cell NE-anchored metric (§4). The full bias accounting and the 7-cell preview predicted-vs-empirical table appear in Appendix B.

4 Trainability: PPO against the NE phase diagram

Protocol. We trained independent PPO runs on each of 37 phase-diagram cells from Figure 1 using the in-repo `JointPPOTrainer` on the `minimal_specialization` scenario, with 50 training iterations of 2048 rollout steps and 4 random seeds per cell (N=148 runs). Two cells (`b0.10_k0.50_c0.50`, `b0.10_k0.90_c0.50`) were skipped in the sweep—the β=0.1, c=0.5 column at $\kappa \in \{0.5, 0.9\}$ was not scheduled—leaving 37 of the 39 Nash-solver cells with PPO data. A subsequent 20-seed noise buy-down (seeds 46–61 added to the original 42–45 at the identical budget; see the buy-down paragraph below) is blended into the committed verdict artifact, so every per-cell statistic reported in this section is a mean over 20 seeds on the 8 buy-down cells and over 4 seeds elsewhere, exactly as committed at `experiments/p3_specialization/phase_diagram_ppo_v2/recalibrated_verdict.json`. Hyper-parameters are pinned in `experiments/p3_specialization/`; the sweep ran on a single 16-core host (`alc-2`) in approximately six hours. The PPO implementation follows the standard recipe of Schulman et al. (2017); the multi-agent joint-policy formulation follows the MAPPO-style centralised-critic setup of Yu et al. (2022). The cells from Appendix B contribute the NE verdict for each (β, κ, c) combination tested.

Per-cell NE-anchored metric. The original v1 sweep scored every cell against a single canonical baseline (the `Random` and `SpecialistPolicy` returns measured at one calibration cell, β=0.25, κ=0.5). That metric was misleading across cells because the random-policy return varies substantially with κ (a low-κ cell is intrinsically harder for any policy, so a small absolute gap against the canonical-cell baseline understates relative PPO success). We replaced it with `gap_closed_ne`, the per-cell gap-closure ratio between Random and the NE *policy profile* recovered for that cell by the heterogeneous Double-Oracle solver (typically 1×`Hero` + 3×`Firefighter` on asymmetric-only and mixed cells, and 4×`Hero` on symmetric-only cells). The per-cell baselines are tabulated in `experiments/nash/phase_diagram/per_cell_baselines.json` and computed by `bucket_brigade/baselines/per_cell.py`; the recipe is the same Random/SpecialistPolicy methodology of Appendix B, applied per cell. The methodological lesson is independent of this paper's specific claims: single-cell baselines mislead cross-cell PPO comparison whenever the random-policy return is itself a function of the cell's parameters, and per-cell calibration is the necessary correction. We report this as a separate contribution alongside the data themselves.

Results. Figure 2 reports the per-cell `gap_closed_ne` (mean±std over seeds; n=20 on the 8 buy-down cells, n=4 elsewhere) across the three c panels. The ordering across NE classes is symmetric_only (0.142; n=11) > mixed (0.096; n=9) > asymmetric_only (0.060; n=11) > no_convergence (−0.024; n=6; the metric falls back to `gap_closed_homogeneous` because no NE policy exists), consistent with the analytical prediction for the three classes the bound covers and placing the empirically-observed mixed class between symmetric_only and asymmetric_only—the location consistent with both pure NE being available but neither being the unique attractor. We report the ordering, not statistical significance: the per-class mean within-cell std ($\approx$ 0.20–0.42) is far larger than the adjacent class-mean separations on `gap_closed_ne` (0.045 and 0.036 between the three NE-bearing classes), so the data is consistent with the predicted ordering but does not reject the null of equal class means under a standard test. A cross-class 4×-budget sweep is in scope for the camera-ready and is the natural significance gate. β-invariance holds at verdict level on all 13 (κ, c) rows of the PPO subgrid with ≥ 2 β samples (the PPO sweep skipped the lone splitting Nash row's β=0.1 *cell*—the one carrying its mixed verdict—so the (κ=0.9, c=0.5) row enters with two asymmetric_only samples); recall from §3 that the environment is bit-identical across β in this extinguish mode (repo issue #442), so cross-β variation in the PPO metric at fixed (κ, c) cannot reflect differences in game dynamics and instead bounds the run-to-run noise of the training pipeline itself – a reading we exploit in the negative-result paragraphs below. PPO recovers substantial value on symmetric-only cells, partial value on mixed and asymmetric-only cells, and approximately none on no-convergence cells, consistent with the hypothesis that PPO converges reliably when a clean symmetric attractor exists, struggles when the only NE requires symmetry-breaking, and fails when no pure-strategy NE is available to converge to.

Is PPO failure on no-convergence cells budget-limited? We ran a 4×-budget sweep on the six no-convergence cells (200 iterations × 4096 rollout steps, 4 seeds, ≈ 6 hours on `alc-6`):[2] mean `gap_closed_homogeneous` worsens to −0.108 (from −0.024 at the original budget). PPO's failure on no-convergence cells is therefore structural, not budget-limited; longer training spends more rollouts in the absence of a pure-strategy attractor and per-seed runs drift further from the cell-specific `SpecialistPolicy` baseline. The load-bearing claim is the direction of the shift; we report the absolute magnitudes as they fall out.

A methodological observation we report as a separate contribution: under the original single-cell baseline of the v1 draft, the 7-cell preview ordering inverted to asymmetric_only (0.262) > symmetric_only (0.091) > no_convergence (−0.176), which would have falsified the §3 prediction. The inversion is an artefact of the metric: asymmetric-only cells at κ=0.9 have a much lower random-policy return than the canonical baseline cell, so PPO's *absolute* improvement looked large despite recovering a smaller fraction of the available NE value. Per-cell baselines resolve the ambiguity and the corrected ordering matches the analytical prediction on both the 7-cell preview and the full 37-cell sweep.

Per-cell trainability predictors: a negative result. The class-level ordering above invites a stronger question: does some scalar property of a cell's NE structure predict how much of the gap PPO closes in *that cell*? The first candidate we proposed—the conditional entropy of the cell's converged heterogeneous NE profile—is retired. Across the 31 cells with a converged NE baseline, per-cell mean conditional entropy against `gap_closed_ne` gives Spearman ρ=0.109 (p=0.56) on the committed artifact (the original all-n=4 computation, preserved at artifact revision `22b1fda6`, gave ρ=0.007, p=0.97; the 20-seed buy-down below re-verified the retirement and the committed headline is the n=20-blend recomputation—insignificant either way); all four entropy aggregates (mean, max, min, spread) are insignificant against both `gap_closed_ne` and `gap_closed_homogeneous`, and the single nominally significant entry (min vs. homogeneous, p=0.038) does not survive Bonferroni correction for the eight tests in the family (α=0.0063). The joined data, statistics, and generating script are committed at `experiments/nash/phase_diagram/entropy_vs_trainability.{py,json,md}`.

The failure is structural, not specific to entropy. Within each (κ, c) column of the grid the converged NE profile—and hence *any* statistic computed from it—is identical across β (indeed the environment itself is bit-identical across β in this extinguish mode; §3), while `gap_closed_ne` varies across β by up to 0.34 within a column (e.g. $-0.32/+0.01/-0.00$ at κ=0.1, c=1.0). The within-column trainability variance is therefore unexplainable by any function of the NE profile, entropy or otherwise. Two further readings follow: per-cell trainability prediction requires inputs beyond the equilibrium profile, and the per-cell std of `gap_closed_ne` reaches 0.99 (median 0.24 across the grid, which is still n=4 outside the 8 bought-down cells)—a noise ceiling that caps the achievable correlation for *any* predictor tested against these means.

[2] The "4×" here counts training iterations (200 vs. 50); because the rollout length also doubles (4096 vs. 2048 steps), the sweep is an 8× multiple in environment steps. The 4×/16× ladder of §5 multiplies iterations at the *fixed* 2048-step rollout, so its labels are true environment-step multiples. Each label matches its committed artifact's configuration.

The noise ceiling, bought down at 20 seeds. A 20-seed noise buy-down on an 8-cell subset spanning the four NE classes (repo issue #443; identical training budget, seeds 46–61 added to the committed 42–45; `experiments/nash/phase_diagram/noise_buydown_precision.md`) quantifies and partially lifts that ceiling. The 95% percentile-bootstrap CI half-widths on per-cell mean `gap_closed_ne` shrink by 2.00–3.64× (median 2.65; 1.84–3.59× on `gap_closed_homogeneous`)—consistent with the $\sqrt{5} \approx 2.24$ expected from 5× the seeds, so the buy-down bought precision, not just seeds. At the higher precision, *none* of the 8 re-measured trainability-ladder verdicts flips (all remain insufficient), and the entropy retirement above survives the recomputation with the n=20 means substituted—the ρ=0.109 (p=0.56) headline quoted there *is* that recomputation, now the committed artifact's own headline (all four aggregates remain insignificant). The n=4 point estimates were noise-dominated but directionally right—the largest per-cell shift is $0.372 \rightarrow 0.043$ at (κ=0.3, c=1.0)—so the class-level conclusions survive the buy-down (the class means reported above already incorporate the n=20 blend) while the original n=4 error bars, as flagged above, did not deserve trust.

The coordination-threshold hypothesis, falsified in its binary form. Our working replacement hypothesis was the coordination threshold k^*: the minimum coalition size whose simultaneous deviation from uniform play yields a positive team-return gap, with the conjecture that PPO—a unilateral best-response method—trains where k^*=1 and fails wherever coordinated multi-agent deviation is required ($k^* \geq 2$). That binary prediction has now been tested in full and is falsified for PPO on this grid. A downstream oracle supplies per-cell k^* for all 75 cells of its (β, κ, c) grid (`rjwalters/thrust#269/#290`; committed verbatim and joined at `experiments/nash/phase_diagram/kstar_vs_trainability.{py,json,md}`). Because k^*, like every other quantity in this extinguish mode, is β-invariant (byte-identical across β within every (κ, c) column), β-replicas are not independent observations: all class tests run at the column level, pooling `gap_closed_ne` across β with n-seed weights. Thirteen of the artifact's 15 effective (κ, c) columns have a PPO sweep and join; two no_convergence columns among them have no NE baseline and drop out of the NE-gap outcome, leaving 11. The pre-registered class test—k^*=1 (3 columns) vs. $k^* \geq 2$ (8 columns) on the pooled NE gap—finds nothing at all: the Mann–Whitney U sits exactly at the null center (rank-biserial 0.00; exact one-sided p=0.539), and the exhaustive 165-assignment permutation gives $\Delta\text{mean} = -0.002$ (two-sided p=0.994). The same split on `gap_closed_homogeneous` (all 13 columns) is likewise insignificant (rank-biserial 0.40, exact one-sided p=0.19). At these class sizes only near-perfect separation is detectable (minimum achievable one-sided p=0.006), so a modest threshold effect could hide—but the observed effect is not modest-and-missed, it is *zero*: PPO closes the NE gap on k^*=2 columns as well as on k^*=1 columns. The literal "unilateral-BR methods fail whenever $k^*>1$" account is dead on this grid; this supersedes the underpowered 2-vs-6 partial-stratification test of the previous revision (p=0.43) and makes k^*'s binary form the second predictor this paper retires.

What survives the falsification, and what it is confounded with. Post hoc—the split was chosen after seeing the primary null, and we flag it as exploratory at every mention—the failure zone is $k^*=k_{\max}=4$: on `gap_closed_homogeneous` (the only gap defined on no_convergence cells) the three k^*=4 columns fall below all ten $k^* \leq 2$ columns (rank-biserial −1.00; exact one-sided p=0.0035, the combinatorial floor at 3 vs. 10; permutation two-sided p=0.021). PPO's cliff is not where coordination is first required (k^*=2) but where the *whole team* must deviate at once. Two structural caveats bound what this can mean. First, κ-confounding: in this artifact k^* is a pure function of κ (κ=0.1→4, 0.3–0.7→2, 0.9→1), so every k^* result—including the failure zone and the correlation below—is observationally equivalent to "κ=0.1 is untrainable"; the coalition mechanism is a candidate *explanation* of the κ effect, not independently identified here. Second, selection bias: no_convergence cells have no NE baseline, so they drop out of the NE-gap join by construction—biased *against* k^* exactly where it makes its strongest prediction. With those flags in place, the comparison to the retired entropy predictor still favors k^*: Spearman k^* vs. gap_closed_homogeneous is $\rho = -0.556$ (p=0.0004, n=37 cells; $\rho = -0.635$, p=0.020 at column level), the only significant pair in either predictor family, against the entropy predictor's n.s. ρ=0.109/0.342. Nor does k^* refine the NE taxonomy: it cross-cuts the verdict classes (every k^* level mixes at least two of them, and asymmetric_only cells appear at all three k^* levels); the single clean alignment is that all no_convergence cells fall in k^*=4 columns, consistent with the post-hoc failure-zone reading—k^* is a coarser, κ-driven trainability marker, not a sharper equilibrium taxonomy. The account must also absorb the earlier awkward datum: a scripted k=1 best response against three frozen uniform opponents already finds a decisive team-return improvement (+13.4/step over uniform, paired 95% CI $[+10.5, +16.6]$, i.e. ≈14%) on the c=0.5 no_convergence cells where PPO stays flat (`experiments/nash/phase_diagram/improvability_oracle.md`), so single-agent improvability alone does not separate trainable from untrainable cells either. This paper accordingly makes no quantitative predictor claim; the retirement of the entropy predictor (re-verified at n=20), the β-invariance impossibility argument, and the falsification of the binary k^* threshold are the results we stand behind—the k^*=$k_{\max}$ failure zone is offered as an exploratory, κ-confounded observation awaiting an out-of-grid test.

Caveats. The original-budget sweep covers 37 of the 39 Nash cells; within-class statistics (n=11 symmetric, n=9 mixed, n=11 asymmetric, n=6 collapse) are wider than the v1 preview but modest for any significance-style claim, hence the explicit ordering-not-significance framing of the Results paragraph above. The PPO budget per cell is modest by recent MARL standards. The gap_closed_homogeneous metric (per-cell Random → SpecialistPolicy × 4) compresses the cross-class separation, and the full four-way ordering is *not* robust to the metric choice: on the committed artifact the homogeneous class means are symmetric (0.041), mixed (0.046), asymmetric (0.030), collapse (−0.024)—the symmetric/mixed pair inverts relative to the NE-anchored ordering, on a 0.005 separation that is well inside the within-class noise. (An earlier revision claimed the homogeneous metric "yields the same qualitative ordering"; that held on the all-n=4 artifact by a 0.0005 margin and inverts under the n=20 blend, so we withdraw it.) What survives on both metrics is the coarser, load-bearing contrast: all three NE-bearing classes sit above no_convergence, and symmetric_only sits above asymmetric_only; the mixed class's exact rank is metric- and noise-sensitive.

5 Social-trap hardness: measured headroom PPO does not capture

The phase-diagram cells of §4 all derive from the `minimal_specialization` scenario, where the equilibrium profile is also a strong team outcome. Bucket Brigade additionally ships a scenario engineered so that the equilibrium is team-suboptimal by construction—a social trap—and on that scenario the benchmark can make a hardness statement that, to our knowledge, no surveyed benchmark makes with committed measurements: *at the standard training budget every trained policy is statistically indistinguishable from random play; at 16× budget exactly one configuration separates from random—marginally, and with no improvement in the mean—while a hand-coded scripted team demonstrates ≈80/step of headroom that no trained configuration approaches.*

The scenario and its anchors. `rest_trap-v1` (frozen scenario ID; 4 agents) has a frozen heterogeneous NE of 3×`FreeRider` + 1×`Firefighter` with team payoff 2984.04/episode, i.e. at

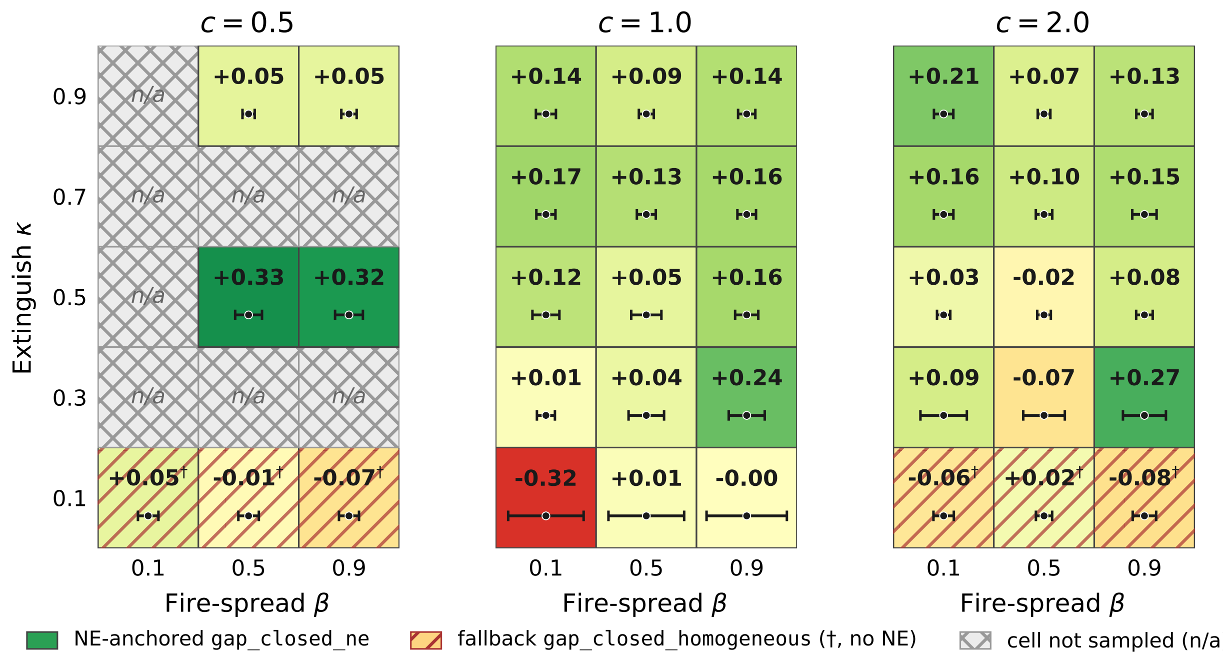

Figure 2: Per-cell PPO `gap_closed_ne` on the 37-cell sweep (50 iterations $\times$ 2048 rollout steps with the in-repo `JointPPOTrainer`; 20 seeds on the 8 buy-down cells, 4 seeds elsewhere, as committed at `experiments/p3_specialization/phase_diagram_ppo_v2/recalibrated_verdict.json`), arranged in three panels by cost $c \in \{0.5, 1.0, 2.0\}$. Mean shown above the Tufte-style horizontal bar; bar half-width is proportional to per-cell std, scaled so the widest std spans 80% of the cell. Symmetric-only cells (top-left of each panel) show the largest gap closure; asymmetric-only cells show positive but smaller gap closure; `mixed` cells fall between; `no_convergence` cells (bottom row; hatched, † marker) report `gap_closed_homogeneous` instead—values are near zero or negative. On `no_convergence` cells the `gap_closed_ne` metric is undefined because no NE policy profile exists; `gap_closed_homogeneous` (per-cell Random $\rightarrow$ SpecialistPolicy $\times$ 4) is the natural fallback because the all-Hero `SpecialistPolicy` is the strongest stationary policy available on those cells, even though it is not a NE. Two cells (β=0.1 at $\kappa \in \{0.5, 0.9\}, c$=0.5, n/a, hatched light grey) were skipped by the sweep schedule. The ordering across NE classes is consistent with the analytical prediction of §3.

most 248.67/step at the 12-night minimum episode length (`bucket_brigade/baselines/release/local/nash/rest_trap-v1.json`). That bound sits *below* the 302.87/step uniform-random baseline: rational play is collectively worse than noise. Two caveats keep this honest. First, per the §3 solver-noise caveat (repo issue #442), the frozen NE *payoff* carries unquantified double-oracle solver noise (a battery-seeded symmetric-oracle retry, below, searched for a better equilibrium and found nothing that stands); the trap *ordering* is nonetheless robust, because the fully scripted `always_rest` team also scores below random (288.55/step, 95% CI $[285.20, 291.65]$). Second, the equilibrium payoff is per-episode while the anchors are per-step; the ≤ 248.67/step bound is conservative in the direction that matters for the trap claim. Table 2 collects the measured anchor ladder.

Symmetric-oracle cycling is a property of the scenario, not a hole in the characterization. Earlier revisions of this paper hedged that the symmetric double-oracle solve for `rest_trap` had never converged, leaving open whether a better (symmetric) equilibrium existed. A battery-seeded retry (repo issue #445; `experiments/nash/rest_trap_seeded_do/RESULTS.md`) settles the question. With the scripted-battery winners—including the 386.60/step specialist—injected into

Anchor	Per-step team reward	Provenance
Frozen NE (3×FR+1×FF)	≤ 248.67	NE payoff 2984.04/ep ÷ 12 nights
`always_rest` ×4 (scripted)	288.55 $[285.20, 291.65]$	scripted battery screen, n=2000
Uniform random	302.87; 302.94 $[301.46, 304.31]$ at n=10k	committed baseline; re-measured upper bound
`het_ppo`, 1× budget (20 seeds)	306.26 $[302.95, 309.33]$	trailing-5 mean, seed-bootstrap 95% CI
`het_ppo`, 16× budget (20 seeds)	307.83 $[305.00, 310.71]$	budget ladder; first `escaped_trap`
`specialist` ×4 (scripted best)	**386.60** $[386.17, 387.03]$	all-scripted battery, n=10k paired

Table 2: Measured anchor ladder on `rest_trap-v1` (per-step team reward). The equilibrium and full resting both score below random—the trap is real—while a hand-coded homogeneous `specialist` team beats uniform random by +83.67/step (paired 95% CI [+82.36, +84.89], n=10,000 paired episodes), demonstrating the scenario is not reward-capped at random-level play. The 16×-budget `het_ppo` row is the lone trained configuration whose CI lower bound clears the random anchor's measured upper bound (305.00 vs. 304.31); see the budget-scaling paragraph for why that flip is variance-driven, not a mean improvement. Sources: `experiments/p3_specialization/scripted_battery/rest_trap.{json,md}`, `experiments/p3_specialization/tier1_runs/tier1_verdict.md`, `experiments/p3_specialization/tier1_runs_trap_escape/`.

the oracle's initial pool, the symmetric double oracle ran its full 50-iteration budget without approaching convergence: minimum per-iteration improvement 11.79 against a threshold of ε=0.01, equilibrium payoff oscillating in $[1611.56, 2316.61]$/episode with no trend—the same basin as the earlier unseeded run—so cycling is not a seeding-coverage artifact; the cooperative region was proposed from iteration 1 and the solver moved away from it. Both candidate profiles the run surfaces are measurably far from equilibrium: the final mixed profile is exploitable by at least 407.50/episode (best deviation `always_rest`—the trap in miniature; rest-leaning deviations gain, work-leaning deviations lose), and the genome image of the scripted specialist team is exploitable at every position (+605 to +870/episode). Nothing found in either run beats the frozen asymmetric NE (3×`FreeRider` + 1×`Firefighter`, 2984.04/episode), which stands as the scenario's characterization: at the measured exploitability bounds `rest_trap` admits no symmetric equilibrium—role differentiation is required—and symmetric-oracle non-convergence is the symmetric-solver view of that fact, now documented as a property of the game rather than an open hole. (The cycling mechanism itself is solver noise: at the verification budget the final solve's "indifferent" support strategies actually score from −258 to +237/episode against the mixture, so each new best response reshuffles the support.) This closes the repository's equilibrium coverage at 12/12 frozen scenarios: 11 converged symmetric equilibria plus `rest_trap` as an explicitly non-converged, exploitability-annotated entry with the asymmetric NE standing.

Why the fraction ladder is inapplicable here—and a vacuous historical pass. The `gap_closed` fraction metric of §4 needs an upper reference that beats its lower reference; on a social trap the equilibrium sits below random, so no valid upper reference exists and the fraction ladder is degenerate (repo issue #434 makes this scenario-aware in the scoring tool: `gap_source=degenerate_reference`, `gap_closed=null`). Before that correction, tier-1 runs on `rest_trap` were scored against `minimal_specialization`-scale references, under which a uniform-random policy alone mapped to `gap_closed` ≈ 6.58 and *any* policy, trained or not, passed the 0.49

gate. Every historical "pass" recorded for `rest_trap` under that gate was therefore vacuous, and no such result—including in earlier internal write-ups of this project—should be read as evidence that a trainer escaped the trap.

Categorical trap-escape verdict. The replacement rule (repo issue #436; `run_tier1_cell.classify_trap_verdict`) computes the percentile bootstrap 95% CI $[lo, hi]$ over seeds of the trained trailing-5 per-step team reward (10,000 resamples, fixed RNG seed) and walks a nested one-sided ladder on lo:

- $lo >$ `scripted_best.ci95_hi`: above_scripted_best;
- $lo >$ `random_ci95_hi`: escaped_trap;
- $lo >$ `ne_bound`: at_random;
- else trapped_at_ne.

The escaped_trap rung deliberately anchors on the random baseline's *measured* 95% upper bound (304.31 at n=10k), not the bare 302.87 point, which carries ±1.4/step measurement noise: a sub-noise clearance of the point is not a statistically supportable "above random" claim.

The honest trained-policy result at the standard budget: at_random. The strongest standard-budget tier-1 result on `rest_trap-v1` is heterogeneous PPO (`het_ppo`, per-agent policies, 20 seeds): trailing-5 mean 306.26, seed-bootstrap CI $[302.95, 309.33] \rightarrow$ verdict at_random. The CI lower bound clears the random *point* by 0.08/step but not the random anchor's own measured upper bound, so `het_ppo` cannot be ruled significantly above random; the quantitative headline is `uplift_over_random`$=+3.39 \pm 7.34$/step. It does clear the NE bound—PPO does not fall *into* the resting trap—while capturing essentially none of the ≈80/step measured scripted headroom. The hypothesis that per-agent policy asymmetry rescues learning on asymmetric-equilibrium scenarios therefore currently has *no* positive interventional evidence on this benchmark (the marginal 16×-budget result below does not supply it either, for the reasons given there); we state this explicitly so that the retracted vacuous gate pass above is not read as such evidence. For context, the tier-1 trainer battery on `minimal_specialization-v1` (`ippo`, `influence`, `hca`, `lola`; 3 seeds each) scores insufficient on the fraction ladder (`gap_closed` mean 0.005–0.125), so the gap is not an artifact of one weak trainer (`experiments/p3_specialization/tier1_runs/tier1_verdict.md`).

Budget scaling: a marginal statistical escape at 16×, with no mean improvement. Is the at-random plateau a training-budget artifact? A pre-registered budget-scaling ladder (repo issue #444; stopping rule: end the ladder at the first escaped_trap) re-ran `het_ppo` and `ippo` at 4× and 16× the tier-1 budget (200/800 iterations × 2048 rollout steps, 20 seeds per cell; `experiments/p3_specialization/tier1_runs_trap_escape/`). Three of the four cells remain at_random (4× `het_ppo` CI lower bound 304.03; 4×/16× `ippo` 301.77/303.46); `het_ppo` at 16× produces the first escaped_trap verdict recorded for any trainer on `rest_trap-v1`: trailing-5 mean 307.83, seed-bootstrap 95% CI $[305.00, 310.71]$, lower bound clearing the 304.31 random-anchor upper bound by +0.69/step. The crossing is robust to the test machinery—the CI lower bound clears the anchor under 300/300 combinations of bootstrap RNG seed and resample count, and a non-bootstrap t-interval on the 20 seed means, $[304.67, 310.99]$, clears it as well—but it must be read for what it is.[3]

[3] Provenance note: the 300/300 RNG/resample-combination sweep and the t-interval trace to the PR #460 judge recomputation as recorded in this paper's committed revision trail (`paper/anvil_pub.bb-workshop.8/changelog.md`,

There is no dose-response on the mean: the plateau is flat across budgets (306.26 → 307.66 → 307.83; all paired same-seed contrasts insignificant, e.g. $16\times - 1\times = +1.57 \pm 4.80$/step, t=1.46), extending the 4× vanilla-PPO flatness of §4 to `het_ppo`, `ippo`, and 16×. *The verdict flip is variance-driven*: extra budget tightens the seed spread (uplift std 8.36 → 6.59), marching the CI lower bound up (302.95 → 304.03 → 305.00) around an essentially unchanged mean—budget buys seed-level consistency, not performance. And *the learnability gap stands*: the mean uplift (+4.96/step) is ≈6% of the measured 83.7/step scripted headroom, and the best single seed anywhere in the ladder captures ≈27% of it, remaining ≈61/step below `scripted_best`. Two caveats travel with the row: the four ladder cells share seed streams, so they are CRN-coupled rather than independent tests, and one of four 95%-CI tests crossing by 0.69/step is weak stand-alone evidence—the verdict rule is pre-registered, so the row stands as scored, as *marginally but significantly above random*, never as "PPO solves the trap at scale." The ladder also supplies the sharpest interventional contrast to date for the per-agent-asymmetry hypothesis—`het_ppo` crosses the escape threshold where `ippo` at the identical budget does not—but the two trainers' means are statistically indistinguishable, so we do not count that as positive evidence either. Net effect: the hardness headline of this section is unchanged and, if anything, sharpened—16× compute buys statistical separation from random and no progress toward the specialist solution.

The resulting benchmark statement is unusually concrete: on `rest_trap-v1` the target is not "do better"; it is a committed, machine-verdicted ladder (≤248.67 < 302.87 < 307.83 < 386.60 per step) in which the `escaped_trap` rung has now been reached exactly once, marginally, by 16×-budget heterogeneous PPO, and the remaining ≈79/step gap to `scripted_best` is known to be realizable by a stationary policy profile. The open challenge is no longer statistical separation from random play; it is capturing headroom.

6 Related work

The MARL community has converged on a handful of canonical benchmarks that trade equilibrium transparency for environment richness. Overcooked-AI (Carroll et al., 2019) is a fully cooperative coordination game on small grid layouts: every Pareto-optimal joint policy is a Nash equilibrium and "the" target is therefore not unique. Melting Pot (Leibo et al., 2021) exposes 88×88×3 RGB observations across more than 50 substrates; its design explicitly does not solve for equilibrium structure on individual substrates. The Hanabi Challenge (Bard et al., 2020) has near-optimal hat-guessing strategies known at 2 players, but the 4–5 player game is open and no algorithm reaches the equilibrium from learning. SMAC and SMACv2 (Ellis et al., 2023; Samvelyan et al., 2019) are cooperative micromanagement against a scripted opponent and "NE" is not a design target. MAgent (Zheng et al., 2018) hosts thousand-agent battles on grids and no published equilibrium analysis exists at scale. PettingZoo's MPE suite (Terry et al., 2021) is the canonical small multi-agent baseline with no equilibrium characterisation across scenarios.

Table 3 compares Bucket Brigade against these on six axes.

The honest comparison is that Bucket Brigade dominates on exactly one column (NE characterisability) and loses on every other (citation footprint, agent count, environment richness, generality). The paper's job is to argue that the NE-characterisability column matters enough to be worth introducing a new benchmark despite the trade. Bucket Brigade does not scale to large populations, does not test "emergent cooperation" in any general sense, and is not a general-purpose MARL benchmark. The niche it fills is the specific question of *equilibrium-convergence quality*.

`paper/anvil_pub.bb-workshop.8/stale_claims_audit.md`), not to a regenerable `experiments/` artifact—weaker than this paper's usual artifact standard, and flagged as such.

Benchmark	Year	Agents	Per-agent $\|\mathcal{A}\|$	NE characterisable?	Coop/comp
Overcooked-AI (Carroll et al., 2019)	2019	2	6	No (any Pareto-opt.)	coop
Melting Pot (Leibo et al., 2021)	2021	2–16	8	No (intractable)	mixed
Hanabi (Bard et al., 2020)	2020	2–5	≤ 20	Partial (2P only)	coop
SMAC/SMACv2 (Ellis et al., 2023; Samvelyan et al., 2019)	2019/23	2–27	$6+n_e$	No	vs. scripted
MAgent (Zheng et al., 2018)	2018	64–10^3	21	No (population)	mixed
PettingZoo MPE (Terry et al., 2021)	2021	2–6	3–5	No	per-scenario
Bucket Brigade (this work)	2026	4 (def.)	8 / 40	**Yes (closed-form)**	**mixed within cell**

Table 3: Bucket Brigade against the six most-cited canonical MARL benchmarks. The right-most column "NE characterisable?" is the only axis on which Bucket Brigade dominates; on every other axis the surveyed benchmarks are richer, more cited, or larger in scope. This is the entire pitch: Bucket Brigade is a *methodological complement*, not a replacement. Per-agent action sizes for Bucket Brigade are 8 at the $H{=}2$ minimal scenario and 40 at the $H{=}10$ default.

The equilibrium-computation and exploitability-evaluation tradition. A second, closer line of prior work evaluates convergence-to-equilibrium directly rather than through benchmark scores. OpenSpiel (Lanctot et al., 2019) ships dozens of small normal-form and extensive-form games together with exact solvers, so "did the learner reach equilibrium?" is answerable there too; the PSRO/NashConv line (Lanctot et al., 2017) made exploitability the standard convergence-to-equilibrium metric for learning in games, and our double-oracle solver and the per-position exploitability bounds of §5 are direct descendants of that machinery. What that tradition evaluates on, however, is a library of *fixed, individually-specified* games—mostly zero-sum or classic matrix games—rather than a single *parametric family* whose equilibrium *structure* (which regime obtains, not just which profile) moves continuously under interpretable environment scalars. The line's own later evaluation work keeps that shape: its benchmark-style move measures populations on repeated rock–paper–scissors, a single fixed game (Lanctot et al., 2023), and its current methodological statement for deep MARL estimates equilibria *empirically* from simulated meta-games (Li and Wellman, 2024) rather than reading them off an analytical map. Likewise the oldest form of NE-ground-truth MARL evaluation—equilibrium *selection* in fixed matrix games whose equilibria are known by inspection, the climbing/penalty-game tradition (Christianos et al., 2022)—supplies ground truth only for individually-specified, stateless two-agent matrices. Bucket Brigade's claim is therefore scoped precisely: it is not the first environment where equilibrium convergence is measurable (that credit belongs to the exploitability tradition), but to our knowledge it is the only published *parametric* cooperative-competitive MARL family in which a closed-form regime map over the parameter space—a phase diagram—supplies the ground truth that a trainability sweep can then be scored against, cell by cell.

7 Discussion and limitations

Bucket Brigade is intentionally small and artificial. The state space is 2304 at the minimal scenario and $\sim 10^{10}$ at the default 10-house scenario, the geometry is a 1-D ring, and the environment ships from a single research repository with no community adoption. We acknowledge each point head-on and defend it as the cost of the equilibrium-transparency property: (a) *Size is the cost of NE transparency.* A richer game with realistic observations, partial observability across many channels, and complex spatial dynamics would lose the property that motivated the work; the state-space cardinality is what makes the analytical reduction in §3 algebraically tractable. Trading richness for tractability is the deliberate design choice. (b) *A custom benchmark is acceptable given the methodological niche.* The question we want to answer—"did this MARL algorithm converge to the analytically-characterised NE?"—cannot be asked of any of the six canonical benchmarks in Table 3. A benchmark designed for this question is therefore a methodological contribution distinct from the contributions those benchmarks make, even if it does not inherit their scale or citation impact. (c) *Bucket Brigade is not a substitute for any of the six.* It does not replace Overcooked for human-AI coordination, Melting Pot for generalisation across substrates, Hanabi for partial-observability cooperation, SMAC for micromanagement against scripted opponents, MAgent for population-scale phenomena, or PettingZoo for the small-multi-agent baseline role. A researcher choosing a benchmark for any of those purposes should not pick Bucket Brigade.

Threats to the contributions. On the analytical side, the closed-form thresholds disagree with empirics by 3–10× in κ-threshold location (§3), and the bound does not predict the empirical mixed class at the higher-c rows. The three identified systematic biases—ring locality, per-agent ownership rewards, and the heuristic-strategy approximation—are not separated by the existing grid; rigorous separation requires a sweep in which β is actually live (the contagion-active phase order or the continuous-extinguish variant of Appendix A.5—in the bernoulli mode used here β is provably inert, so no ρ-sweep can activate it; repo issue #442), and a ring-Markov refinement of the survival coefficient A to predict the mixed boundary in closed form. Whether the solver's converged asymmetric profiles are themselves ε-NE under common-random-number re-evaluation is an open thread (repo issue #459). On the empirical side, per-class sample size is modest ($n \in \{6, 9, 11, 11\}$) and the budget per run is modest; the 4×-budget sweep rules out one budget-limited hypothesis for the no-convergence row but not the cross-class question, which is in scope for the camera-ready, and the 20-seed noise buy-down of §4 lifts the per-cell seed-noise ceiling on 8 of the 37 cells (verdicts unchanged) without yet covering the rest of the grid. The k^* join of §4 inherits two limits of its own: k^* is a pure function of κ on this grid, so no within-grid test can separate the coalition mechanism from a plain κ effect (the exploratory $k^*{=}k_{\max}$ failure-zone reading needs cells where k^* varies within κ, which the current grid does not contain), and the class tests carry a power floor (minimum achievable one-sided $p{=}0.006$ at 3 vs. 8 columns) that only near-perfect separation could beat—the primary null is a measured zero effect, not merely a failure to reach significance. The per-cell-baseline methodological result of §4 stands independently of both the longer-budget outcome and the ring-Markov refinement: it is load-bearing on cross-cell PPO comparison whenever the random-policy return varies across cells, regardless of the analytical reduction's fate.

Reproducibility. The environment is shipped as a pip-installable Python package (`pip install bucket-brigade`); the figure-by-figure reproduction recipe lives in the in-repo `docs/PAPER_RESULTS.md`. The per-cell baselines used by the §4 metric are produced by `bucket_brigade/baselines/per_cell.py` and serialised at `experiments/nash/phase_diagram/per_cell_baselines.json`; both ship with the repository so the recalibrated `gap_closed_ne` column of Figure 2 can be re-derived from the raw PPO-sweep checkpoints. (The v2 recalibrated artifacts this paper reports reproduce byte-for-byte from the committed per-seed summaries, including the seed-46–61 buy-down summaries blended into the 8 extended cells—every §4 statistic and the rendered Figure 2 match the artifacts at repository HEAD; a reproduction drift in the *superseded* v1 tables is tracked openly as repo issue #461.)

Three pieces of infrastructure added after the sweeps target the failure mode benchmark consumers actually hit. *Frozen scenario IDs*: every scenario this paper reports on carries a versioned, immutable registry entry (`minimal_specialization-v1`, `rest_trap-v1`, and the registered phase-diagram cells such as `asym_b05_k09_c05-v1`), guarded by bit-parity regression tests (`bucket_brigade/envs/registry.py`; repo PR #441). Numbers in this paper cite frozen IDs, never bare scenario names. *Parity manifest and check*: a committed manifest records, per frozen scenario, the canonical uniform-random per-step team reward with its n=1000 bootstrap CI and a scenario-parameter fingerprint; a seconds-cheap CLI (`python -m bucket_brigade.baselines.parity --all`) re-measures an installed build against the manifest with a CI-derived tolerance and fails loudly on mismatch (`docs/PARITY.md`; repo PR #439). *The motivating incident*: a downstream harness ran a full study of this benchmark on a differently-weighted reward configuration whose returns were $\sim 7\times$ repo-native scale; nothing failed loudly, and the mismatch was only exposed by a dedicated repo-side scripted oracle (repo PR #432). Results produced off-scale are worse than no results; the parity check converts that class of silent mismatch into an immediate failure, and we ask anyone publishing numbers about Bucket Brigade to cite the frozen scenario ID, the manifest version, and a passing parity check.

Frozen baseline checkpoints (the analytical `Hero`, `Firefighter`, and `FreeRider` agents used to anchor the phase-diagram solver and the per-cell PPO metric) will be hosted on the HuggingFace Hub; the publication pathway for the baselines is in progress and not yet complete at the time of this draft. The source repository is at https://github.com/rjwalters/bucket-brigade.

Conclusion

We presented Bucket Brigade, a parametric cooperative-competitive MARL benchmark whose Nash-equilibrium regime map is derivable in closed form and testable cell by cell, and everything the paper establishes is scored against that map: the closed-form bound predicts the qualitative phase order that both the 39-cell empirical diagram and the 37-cell PPO sweep recover (the latter under the per-cell NE-anchored metric of §4; the bound's quantitative κ-thresholds miss by 3–10$\times$ and a fourth `mixed` class escapes it, both reported as part of the claim); two intuitive per-cell trainability predictors are retired in sequence; and on `rest_trap-v1` trained policies capture essentially none of a machine-verdicted $\approx$80/step scripted headroom even at 16$\times$ budget. The open challenge is equally concrete: predict the `mixed` boundary in closed form, find a per-cell trainability predictor that survives an out-of-grid test, and capture the committed `rest_trap` headroom—each answerable, cell by cell, against a closed-form regime map that to our knowledge no other published MARL benchmark family pairs with its training grid. Bucket Brigade is a methodological complement to the canonical MARL benchmarks for the specific question of equilibrium-convergence quality, not a replacement for any of them.

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A Formal environment specification

This appendix gives the self-contained mathematical contract for the Bucket Brigade environment. §2 of the body is a compressed summary; the material below is what a reader needs to reimplement the environment from scratch in any language, with the seven-phase ordering, the independent-workers extinguish formula, and the per-agent reward decomposition all spelled out. The notation matches §2 verbatim.

A.1 Players and parameters

Player set. A finite set of agents $\mathcal{I} = \{1, 2, \ldots, N\}$, with $N \geq 1$ fixed across an episode. Agents are *positionally symmetric*: the environment dynamics are invariant under any permutation $\sigma : \mathcal{I} \to \mathcal{I}$ applied jointly to every per-agent quantity (locations, signals, actions, rewards). The asymmetric reward variants of §A.5 (per-agent home-position cost, per-agent ownership weights) break this symmetry only when their dedicated parameters are nonzero; the symmetric default game preserves exchangeability.

Topology. The world is a cycle graph $\mathcal{H} = \mathbb{Z}/H\mathbb{Z}$ with $H \geq 2$ vertices ("houses") and edge set $\{(h, h{+}1 \bmod H) : h \in \mathcal{H}\}$. House indices are taken mod H throughout. The two neighbours of house h are $h{-}1 \bmod H$ and $h{+}1 \bmod H$.

Parameter family. The full parameter family is $\theta = (\beta, \kappa, \rho, c, H, N, T_{\min}, W)$ where $\beta \in [0, 1]$ is the per-night, per-neighbour fire-spread probability; $\kappa \in [0, 1]$ is the per-agent solo extinguish probability; $\rho \in [0, 1]$ is the per-night, per-Safe-house spontaneous-ignition probability; $c \in \mathbb{R}_{\geq 0}$ is the per-night work cost; $H \in \mathbb{Z}_{\geq 2}$ is the number of houses; $N \in \mathbb{Z}_{\geq 1}$ is the number of agents; $T_{\min} \in \mathbb{Z}_{\geq 0}$ is the minimum nights before termination is allowed; and W is the reward tuple of §A.4.

The default "10-house" scenario uses $H{=}10, N{=}4, \beta{=}0.25, \kappa{=}0.5, \rho{=}0.02, c{=}0.5, T_{\min}{=}12$ and reward weights $W = (R_{\text{team}}, P_{\text{team}}, r_{\text{own}}, r_{\text{other}}, p_{\text{own}}, p_{\text{other}}) = (100, 100, 20, 0, 40, 0)$. The minimal parameterisation sets $H{=}2, N{=}4$.

A.2 State, action, observation

State. The environment state at the start of night t is

$$s_t = (h_t, \ell_t, \zeta_t, t) \in \{S, B, R\}^H \times \mathcal{H}^N \times \{0, 1\}^N \times \mathbb{N},$$

where $h_t \in \{S, B, R\}^H$ is the vector of house statuses (Safe, Burning, Ruined), $\ell_t \in \mathcal{H}^N$ is the vector of agent locations, $\zeta_t \in \{0, 1\}^N$ is the vector of one-bit broadcast signals emitted on the previous night, and $t \in \mathbb{N}$ is the night counter. The Ruined status is absorbing; Safe and Burning are mutually reachable.

Initial state. At $t{=}0$ each house is independently Burning with probability ρ and Safe otherwise; agent locations are $\ell_{0,i}{=}0$ and signals $\zeta_{0,i}{=}0$ for all i.

State cardinality. $|\{S,B,R\}^H \times \mathcal{H}^N \times \{0,1\}^N| = 3^H \cdot H^N \cdot 2^N$. For $H{=}2, N{=}4$ this is $9 \cdot 16 \cdot 16 = 2304$ states; for $H{=}10, N{=}4$ it is $\approx 9.44 \times 10^9$. The state space is polynomial in (H, N) rather than exponential in observation pixels.

Action. Each agent i chooses an action $a_{t,i} = (a^{\text{house}}_{t,i}, a^{\text{mode}}_{t,i}, a^{\text{sig}}_{t,i})$ from the product set $\mathcal{A} = \mathcal{H} \times \{0,1\} \times \{0,1\}$, with $a^{\text{house}} \in \mathcal{H}$ the target house index, $a^{\text{mode}} \in \{0,1\}$ the Work bit (1 = Work, 0 = Rest), and $a^{\text{sig}} \in \{0,1\}$ the broadcast bit emitted this night and observable to all agents on the next. Per-agent cardinality is $|\mathcal{A}| = 4H$ (8 at $H{=}2$, 40 at $H{=}10$); joint-action cardinality is $|\mathcal{A}^N| = (4H)^N$ (4096 at $H{=}2, N{=}4$). The signal bit is strictly informational, incurs no cost, and does not constrain the agent's mode or location; signals support cheap-talk equilibria but the interpretation is not part of the specification.

Observation. The environment is perfect-monitoring with one-step delay on signals: at the start of night t, every agent observes the same tuple $o_t = (h_t, \ell_{t-1}, \zeta_{t-1}, \alpha_{t-1}, t)$, where $\alpha_{t-1} \in \mathcal{H}^N \times \{0,1\}^N$ is the previous night's joint (house, mode) slice. There is no partial observability of houses or agent positions and no private observations.

A.3 Transition dynamics

A single night consists of seven sequential phases applied to the state. The phase order is load-bearing: swapping any two adjacent phases changes the transition kernel. The order is (1) signal write, (2) location and mode write, (3) extinguish, (4) burn-out, (5) spread, (6) spontaneous ignition, (7) reward and termination check. Write $w_h(a) := |\{i : a_i^{\text{house}} = h \wedge a_i^{\text{mode}} = 1\}|$ for the number of agents who Work at house h under joint action a.

Signal and bookkeeping (phases 1–2). $\zeta_t \leftarrow (a^{\text{sig}}_{t,i})_{i \in \mathcal{I}}$, $\ell_t \leftarrow (a^{\text{house}}_{t,i})_{i \in \mathcal{I}}$. These updates do not alter house states.

Extinguish phase (phase 3). For each house h with $h_{t,h} = B$, draw $U \sim \text{Uniform}[0,1]$ and update

$$h^{(3)}_{t,h} = \begin{cases} S & \text{if } U < 1 - (1-\kappa)^{w_h(a_t)} \\ B & \text{otherwise} \end{cases}$$

The independent-workers extinguish formula: each Worker at the house independently succeeds with probability κ, so at least one succeeds with probability $1 - (1-\kappa)^{w_h}$. Safe and Ruined houses are unaffected.

Burn-out phase (phase 4). Every house that survived the extinguish phase as Burning transitions deterministically to Ruined: $h^{(4)}_{t,h} = R$ if $h^{(3)}_{t,h} = B$, else $h^{(4)}_{t,h} = h^{(3)}_{t,h}$.

Spread phase (phase 5). For each pair (h, h') with $h^{(4)}_{t,h} = B$, $h' \sim h$ on the ring, and $h^{(4)}_{t,h'} = S$, draw independent $U \sim \text{Uniform}[0,1]$ and set $h^{(5)}_{t,h'} = B$ if $U < \beta$. Under the canonical phase order, phase 4 has already cleared every Burning source house to Ruined, so phase 5 is inert in the canonical mode and the only path Safe $\to$ Burning during a night is the spontaneous-ignition phase 6. A contagion-active variant swaps phases 4 and 5; under that variant a Safe house with two Burning neighbours has combined ignition probability $1 - (1-\beta)^2$. The phase-diagram analysis in the body uses the canonical order.

Spontaneous ignition phase (phase 6). For each house h with $h^{(5)}_{t,h} = S$, draw independent $U \sim \text{Uniform}[0,1]$ and set $h^{(6)}_{t,h} = B$ if $U < \rho$. Ruined houses are unaffected; freshly-Ruined houses from phase 4 cannot re-ignite. The resulting state is $h_{t+1} := h^{(6)}_{t}$.

A.4 Reward structure

Rewards are paid at the end of each night, after all transition phases. The per-agent reward $r_{t,i}$ decomposes into three independently parameterised terms.

Work/rest term. $r^{\text{work}}_{t,i} = -c$ if $a^{\text{mode}}_{t,i} = 1$ (Work), and $+c_{\text{rest}}$ if $a^{\text{mode}}_{t,i} = 0$ (Rest), with $c_{\text{rest}} = 0.5$ as the implementation default. Rest is a positive payoff that opt-in Work forgoes; the free-rider gradient is the $c + c_{\text{rest}}$ asymmetry.

Per-house ownership term. Each house h has a designated owner $\text{owner}(h) \in \mathcal{I}$ assigned at episode start by round-robin: $\text{owner}(h) = h \bmod N$. The per-house contribution to agent i's reward is

$$r^{\text{own}}_{t,i,h} = \mathbb{1}[h_{t-1,h} \neq S \wedge h_{t+1,h} = S] \cdot w^{\text{save}}_{i}(h) - \mathbb{1}[h_{t+1,h} = R] \cdot w^{\text{ruin}}_{i}(h),$$

with weights

$$w^{\text{save}}_{i}(h) = \begin{cases} r_{\text{own},i} & i = \text{owner}(h) \\ r_{\text{other},i} & \text{otherwise} \end{cases}, \qquad w^{\text{ruin}}_{i}(h) = \begin{cases} p_{\text{own},i} & i = \text{owner}(h) \\ p_{\text{other},i} & \text{otherwise.} \end{cases}$$

The save event fires once on the night the house transitions from non-Safe to Safe; the ruin penalty fires every night the house remains Ruined, so a ruined house imposes a recurring cost until the episode terminates.

Team-welfare term. Let $n_S(t) := |\{h : h_{t+1,h} = S\}|$ and $n_R(t) := |\{h : h_{t+1,h} = R\}|$. The team term, paid to every agent equally, is $r^{\text{team}}_{t} = R_{\text{team}} \cdot n_S(t)/H - P_{\text{team}} \cdot n_R(t)/H$.

Total reward. $r_{t,i} = r^{\text{work}}_{t,i} + \sum_{h \in \mathcal{H}} r^{\text{own}}_{t,i,h} + r^{\text{team}}_{t}$.

A.5 Horizon, information, symmetries, variants

Termination. The episode terminates at night T when $t \geq T_{\min}$ *and* at least one of the following holds: all houses are Safe, all houses are Ruined, or no house is Burning. Under canonical parameters with $\rho > 0$ the no-fires-burning condition holds only transiently, so termination typically occurs via the absorbing all-Ruined condition or via a transient Safe state that survives the spontaneous-ignition draw. Expected episode length is finite for $\rho > 0$, but the game is not strictly finite-horizon.

Information structure. The information structure is symmetric perfect monitoring: every agent observes the full state h_t, the previous joint location/mode/signal $(\ell_{t-1}, \alpha^{\text{mode}}_{t-1}, \zeta_{t-1})$, and the night counter t. No agent has private information about house states or other agents' actions. The signal channel is non-binding cheap talk: $\zeta_{t,i}$ is observable to every agent at night $t+1$ but is functionally decoupled from $a^{\text{mode}}_{t+1,i}$.

Symmetry properties. The default game enjoys agent exchangeability (the transition kernel and reward function are invariant under any permutation σ of agent indices) and ring rotation (the transition kernel is invariant under any rotation $\tau : h \mapsto h + k \bmod H$ applied jointly to $h_t, \ell_t, a_t^{\text{house}}$; reward invariance requires ownership weights to be permuted consistently with τ). Asymmetric reward variants break agent exchangeability; the position-constrained variant breaks ring rotation by anchoring each agent to a home position.

Optional dynamic variants. Four optional variants are part of the calibrated parameter family, each gated by a parameter that defaults to the no-op value: (a) adjacent-only action validity (per-agent home position h_i^{home} with a ring-distance-$\leq$1 constraint on a_i^{house}); (b) continuous-extinguish (each Work at a Burning house adds increment σ to a per-house accumulator; the fire transitions to Safe deterministically when the accumulator reaches 1; fires do not burn out); (c) two-phase signalling (each night becomes two micro-rounds with round-1 emitting signal only and round-2 observing everyone's round-1 signal); (d) distance-cost asymmetry (work cost augmented to $c + \alpha \cdot d(h_i^{\text{home}}, a_i^{\text{house}})$ where d is ring-arc distance, breaking ring-rotation symmetry). These variants are out of scope for the paper but are part of the broader calibrated family.

A.6 Relationship to canonical game-theory templates

Bucket Brigade recovers, as limiting cases or restrictions, several textbook templates.

Volunteer's Dilemma. At $N{=}1, H{=}1$ with $\rho > 0, \beta{=}0$, each night the single agent faces a binary choice: Work (pay c, save the house with probability κ) or Rest (gain c_{rest}, lose the house in the next phase). This is the textbook single-shot Volunteer's Dilemma (Diekmann, 1985). The N-player generalisation at $N > 1, H = 1$ is the *N-player Volunteer's Dilemma*: the team needs at least one volunteer per fire, with the free-rider incentive equal to the $c + c_{\text{rest}}$ gap.

N-Player Public Goods Game. With $r_{\text{own}} = r_{\text{other}}$ and $p_{\text{own}} = p_{\text{other}}$ (ownership weights collapsed to uniform), the per-house ownership term reduces to a non-excludable common good and the team term is paid equally regardless of contribution. The marginal private return to Work is then $\kappa \cdot (R_{\text{team}} + P_{\text{team}})/H - c$, which can be strictly less than the social marginal return $\kappa \cdot (R_{\text{team}} + P_{\text{team}}) - c$, producing the textbook public-goods underinvestment gradient (Ledyard, 1995).

Stag Hunt. In the limit $\kappa \to 1$ a single worker is sufficient to extinguish any fire, but the work cost c remains. The game then has two natural NE: the all-stag cooperative NE where every agent works coordinated houses (high payoff, requires coordination beliefs) and the all-hare lazy NE where everyone Rest-defects (low payoff, no coordination required)—the canonical Stag Hunt payoff structure (Skyrms, 2003). The intermediate-κ regime interpolates between Stag Hunt and the public-goods regime.

Free-rider problem (rest-trap). At specific parameter cells the unique pure-strategy NE is asymmetric: $N{-}1$ agents play Rest and one agent plays full-force Work. This is the textbook free-rider problem in the bucket-brigade vocabulary; the phase diagram across (β, κ, c) shows the boundary between symmetric-NE-dominant and asymmetric-NE-dominant regimes.

Stochastic Game. Stripped to its bones, Bucket Brigade is a finite-state, finite-action, discounted stochastic game with N players, symmetric perfect monitoring, and reward decomposition into private and team components. Every formal result for finite stochastic games (Shapley, 1953)—minimax, equilibrium existence in stationary mixed strategies, value iteration convergence—applies directly. The contribution of the Bucket Brigade design is not in the game-theoretic class but in the *size*: at the minimal parameterisation the game is small enough that standard tools actually compute, which is the property the paper exploits.

A.7 Notation summary

Symbol	Meaning
$\mathcal{I} = \{1, \ldots, N\}$	Agent set; N = population size
$\mathcal{H} = \mathbb{Z}/H\mathbb{Z}$	House ring; H = number of houses
S, B, R	House statuses: SAFE, BURNING, RUINED (R absorbing)
β	Per-night, per-neighbour fire-spread probability
κ	Solo-agent extinguish probability (independent-workers model)
ρ	Per-night, per-SAFE-house spontaneous-ignition probability
c	Per-night work cost (paid when $a_i^{\text{mode}} = 1$)
c_{rest}	Per-night rest payoff (implementation default 0.5)
$T_{\min}$	Minimum nights before termination allowed
$a_{t,i} = (a^{\text{house}}, a^{\text{mode}}, a^{\text{sig}})$	Per-agent action: target house, work bit, signal bit
$w_h(a)$	Number of WORKers at house h under joint action a
$R_{\text{team}}, P_{\text{team}}$	Team reward / penalty per fraction-safe / fraction-ruined
$r_{\text{own}}, r_{\text{other}}$	Per-agent save-reward weights (own / other-owned house)
$p_{\text{own}}, p_{\text{other}}$	Per-agent ruin-penalty weights (own / other-owned house)
$n_S(t), n_R(t)$	Counts of SAFE / RUINED houses at end of night t
$\zeta_t \in \{0,1\}^N$	Vector of broadcast signals (observable next night)
$\ell_t \in \mathcal{H}^N$	Vector of agent locations after phase 2

Table 4: Notation summary. Every downstream section (§3 equilibrium analysis, §4 experiments, §7 discussion, Appendix B NE derivation) reuses this vocabulary without redefinition.

Reproducibility note. The seven-phase ordering of §A.3 is load-bearing: implementers must apply phases in the exact order (signal write, location/mode write, extinguish, burn-out, spread, spontaneous ignition, reward). The independent-workers extinguish formula of §A.3 and the post-extinguish, pre-burn-out spread test are the two details most often misread on first encounter; both are explicit above to make reimplementation tractable. A reader who reimplements from this appendix alone—using any RNG and any programming language—should obtain matching equilibrium structure and policy-evaluation results at the documented parameter cells.

B Analytical NE characterisation for the 4-agent game

This appendix gives the full derivation of the closed-form boundary inequalities reported in §3 and the predicted-vs-empirical comparison table referenced in Figures 1 and 2. The derivation reduces the H=10 ring with $T_{\min}$=12 to a one-house, four-agent, single-night stage game whose payoffs absorb multi-step contagion and ignition dynamics into a single survival coefficient. The three NE structures fall out as algebraic inequalities in $(\kappa, \tilde{A}, c_{\text{gap}})$.

B.1 Setup

The analysis is restricted to the `minimal_specialization` parameter cell with H=10, N=4, $T_{\min}$=12 and reward tuple $W = (R_{\text{team}}, P_{\text{team}}, r_{\text{own}}, r_{\text{other}}, p_{\text{own}}, p_{\text{other}}) = (10, 10, 50, 0, 100, 0)$ at ρ=0.02. The free parameters are (β, κ, c).

The analytical reduction is *not* a derivation of the full stochastic-game subgame-perfect equilibrium. The full game's NE conditions exist only via numerical value iteration on the $\geq 9 \times 10^9$-state space, which does not generalise across (β, κ, c) in closed form. Restricting to the per-night stage game under stationary opponent strategies is the load-bearing simplification that makes algebraic boundary inequalities possible; §B.5 reports where this restriction produces visibly wrong predictions.

Mapping to the empirical strategy space. The empirical phase diagram is computed by the heterogeneous Double-Oracle solver over continuous 10-D heuristic-parameter vectors, not the binary WORK/REST stage game the derivation uses. The bridge is the work-tendency parameter: `Hero` (`work_tendency`=1.0) maps to analytical all-WORK; `Firefighter` (`work_tendency`=0.9) is treated as near all-WORK; `FreeRider` (`work_tendency`=0.2, `rest_reward_bias`=0.9) maps to all-REST. Empirical verdicts map to analytical NE structures as: symmetric_only with profile 4×`Hero` → symmetric all-WORK; asymmetric_only with profile 1×`Hero` + 3×`Firefighter` → asymmetric 1-WORKER; no_convergence → no-pure-NE collapse. The mapping is approximate—`Firefighter`s at `work_tendency`=0.9 are not literal REST agents—and the quantitative bias this introduces is named in §B.5.

Single-house mean-field reduction. Instead of analysing the full 10-house ring with 4 agents distributed across houses, we collapse the analysis to a representative single house with all 4 agents present, where the night's outcome is paid out as the expected per-agent payoff under the stage-game profile. Per-night reward decomposition follows §A.4: a work cost $-c$ on WORK and rest payoff $+c_{\text{rest}}$=0.5 on REST; a per-house ownership term weighted by r_{own}=50 on save and p_{own}=100 on every RUINED night; and a team term scaled by R_{team}=P_{team}=10. In the single-house reduction ownership accrues to the house's designated owner and is shared equally across agents in the steady-state assumption that each agent owns one house and the rotation is symmetric.

B.2 Mean-field stage-game reduction

Survival coefficient. With $k \in \{0, 1, 2, 3, 4\}$ workers at the house on a given night, the probability of extinguish in phase 3 is $1 - (1 - \kappa)^k$. Under the canonical phase order the spread phase has no BURNING source houses to spread from, so β does not enter the leading-order single-night payoff and contagion enters only via spontaneous ignition at rate ρ per SAFE house. Let s be the probability a house is SAFE; conditional on SAFE at night t and k workers on night t+1, the per-night ruin probability is

$$q(k) := 1 - \Pr[s_{t+1}{=}S \mid s_t{=}S, k] = \rho \cdot (1 - \kappa)^k.$$

Over $T_{\min}$=12 nights the probability the house survives is approximately $(1-q(k))^{T_{\min}} \approx 1-T_{\min} \cdot q(k)$ for small $q(k)$. Collecting the $q(k)$-dependent terms of the expected per-house reward yields

$$\mathbb{E}[\text{per-house reward loss} \mid k] = A \cdot \rho \cdot (1 - \kappa)^k,$$

with

$$A = r_{\text{own}} \cdot T_{\min} + p_{\text{own}} \cdot T_{\min} + P_{\text{team}} \cdot T_{\min}/H = 50 \cdot 12 + 100 \cdot 12 + 10 \cdot 12/10 = 1812.$$

The marginal benefit of adding the k-th worker (going from k to $k+1$) is the reduction in expected loss,

$$\Delta(k \to k+1) = A \cdot \rho \cdot [(1-\kappa)^k - (1-\kappa)^{k+1}] = A \cdot \rho \cdot \kappa \cdot (1-\kappa)^k.$$

Writing $\tilde{A} := A\rho = 1812 \cdot 0.02 = 36.24$ for the per-night survival coefficient, the marginal benefit at the k-th worker is $\tilde{A} \cdot \kappa \cdot (1-\kappa)^k$.

Stage-game payoffs. Each agent picks $a^{\text{mode}} \in \{0, 1\}$. With k_{-i} other workers under the joint profile, agent i's per-night payoff difference between Work and Rest is

$$u_i(\text{Work} \mid k_{-i}) - u_i(\text{Rest}) = \tilde{A} \cdot \kappa \cdot (1-\kappa)^{k_{-i}} - c_{\text{gap}},$$

with $c_{\text{gap}} := c + c_{\text{rest}}$. For the phase-diagram driver's c=0.5 and the default c_{rest}=0.5, $c_{\text{gap}} = 1.0$.

Ring locality (statement). The single-house reduction assumes all 4 agents are at the same house. The empirical setup spreads 4 agents across 10 houses each night; under a symmetric uniform policy, each house sees $E[k] = N/H = 0.4$ workers per night. The derivation in §B.3 keeps the single-house framing (all 4 agents at one fire) because that is the case that determines whether a "1 Worker, 3 free-riders" profile is stable; the ring-locality correction is the perturbation in §B.5.

B.3 The three NE candidates

Symmetric all-Work NE. Under the all-Work profile, every agent plays a_i = Work. For this to be a NE, a unilateral deviation to Rest must not strictly increase payoff. The deviation comparison has k_{-i}=3:

$$\tilde{A} \cdot \kappa \cdot (1-\kappa)^3 \geq c_{\text{gap}} \qquad \text{(S)}.$$

Substituting $\tilde{A}$=36.24, c_{gap}=1.0: $f(\kappa) := \kappa(1-\kappa)^3$ is unimodal on $[0, 1]$ with maximum $f(1/4) = 27/256 \approx 0.105$. Solving $f(\kappa) = 0.0276$ numerically gives roots $\kappa \approx 0.030$ and $\kappa \approx 0.65$; the symmetric NE exists for $\kappa \in [0.030, 0.65]$.

Asymmetric 1-Worker NE. Under the asymmetric profile exactly 1 agent Works and 3 Rest. Two conditions: the lone Worker prefers Work (k_{-i}=0) and each free-rider prefers Rest (k_{-i}=1):

$$\tilde{A} \cdot \kappa \geq c_{\text{gap}} \quad \text{AND} \quad \tilde{A} \cdot \kappa \cdot (1-\kappa) < c_{\text{gap}} \qquad \text{(A)}.$$

Condition (a): $36.24 \cdot \kappa \geq 1.0 \Leftrightarrow \kappa \geq 0.028$. Condition (b): $g(\kappa) := 36.24\kappa(1-\kappa)$ is unimodal with $g(1/2) = 9.06$; the roots of $g(\kappa) = 1.0$ are $\kappa \approx 0.028$ (lower) and $\kappa \approx 0.972$ (upper). Condition (b) requires κ outside $[0.028, 0.972]$. The intersection of (a) and (b) is essentially empty below κ=0.028 and gives the high-κ band $\kappa \gtrsim 0.972$, the Stag-Hunt-like regime where a single worker is overwhelmingly likely to extinguish and the free-rider's marginal contribution is negligible.

No-pure-NE collapse. The all-Rest profile is never a pure-strategy NE for any $\tilde{A} > 0$ such that $\tilde{A}\kappa > c_{\text{gap}}$, because a single deviation to Work earns $\tilde{A}\kappa - c_{\text{gap}} > 0$. The no-pure-NE collapse regime is the parameter regime where neither (S) nor (A) holds AND the all-Work profile is dominated. From the above, $\tilde{A}\kappa < c_{\text{gap}}$ implies the lone-Worker deviation does not pay (A (a) fails) and a fortiori (S) fails (since $(1-\kappa)^3 < 1$ for $\kappa > 0$):

$$\tilde{A} \cdot \kappa < c_{\text{gap}} \implies \text{no pure-strategy NE (collapse)} \qquad \text{(C)}.$$

κ range	Symmetric (S)	Asymmetric (A)	Collapse (C)	Predicted verdict
$\kappa < 0.028$	No	No	Yes	collapse (no-pure-NE)
$0.028 \leq \kappa \leq 0.030$	Borderline	Empty	No	borderline; may be mixed
$0.030 \leq \kappa \leq 0.65$	Yes	No	No	symmetric_only
$0.65 < \kappa \leq 0.972$	No	No	No	mixed or transition
$\kappa > 0.972$	No	Yes	No	asymmetric_only

Table 5: Predicted NE structure under the mean-field reduction at c=0.5, ρ=0.02, $T_{\min}$=12. The qualitative phase order from κ-small to κ-large is collapse → symmetric_only → mixed/no-pure → asymmetric_only, matching the empirical ordering on the 39-cell grid at c=0.5. β does not enter under the canonical phase order.

Summary table. For the `minimal_specialization` parameter cell at c=0.5 (so c_{gap}=1.0), ρ=0.02, $T_{\min}$=12, and $\tilde{A} = 36.24$:

B.4 Predicted vs. empirical phase table (7-cell preview)

Before the full 39-cell grid of §3 was computed, the reduction was first checked against a 7-cell preview at c=0.5, ρ=0.02—the historical record matters because the preview is where the quantitative threshold disagreements were first observed and owned, and because the single-cell-baseline inversion of §4 was diagnosed on the same preview. Table 6 reproduces that comparison; the biases it exposes are dissected in §B.5 below.

β	κ	Empirical verdict	Predicted verdict	Match?
0.10	0.10	no_convergence	symmetric_only (in 0.030–0.65 band)	×
0.50	0.10	no_convergence	symmetric_only	×
0.90	0.10	no_convergence	symmetric_only	×
0.50	0.50	symmetric_only	symmetric_only	✓
0.90	0.50	symmetric_only	symmetric_only	✓
0.50	0.90	asymmetric_only	mixed/borderline (predicted threshold 0.972)	partial
0.90	0.90	asymmetric_only	mixed/borderline	partial

Table 6: Predicted vs. empirical NE verdict on the 7-cell preview reported in §3 of the body. Qualitative-vs-quantitative split: β-independence holds 3/3 on κ-rows with both β values; the qualitative phase ladder no_convergence → symmetric_only → asymmetric_only matches; exact cell-by-cell agreement is 2/7 because the predicted κ thresholds are wrong. The predicted collapse threshold κ=0.028 is 3–4× too low; the predicted asymmetric-NE threshold κ=0.972 is $\sim$ 10% too high. The full 39-cell grid in §3 of the body widens the sample without changing the qualitative findings.

B.5 Where the reduction breaks

Six sources of systematic bias are identifiable from the derivation.

Ring locality (the H=10 → mean-field gap). The reduction puts all 4 agents at one house. Empirically, 4 agents distribute across 10 houses each night; under a symmetric uniform policy, each house sees $E[k] = 0.4$ workers per night on average rather than 4. The effective per-house extinguish probability is $1 - (1-\kappa)^{0.4}$ rather than $1 - (1-\kappa)^4$—a much weaker defence. For κ=0.5: full-mean-field gives $1 - 0.5^4 = 0.9375$; the ring-corrected average gives $1 - 0.5^{0.4} \approx 0.24$. The ring-corrected $\tilde{A}$ is therefore much smaller than 36.24, and the collapse boundary κ at which $\tilde{A}\kappa < c_{\text{gap}}$ shifts substantially upward. Predicted correction: the collapse threshold moves from

κ=0.028 (single-house) to roughly $\kappa \in [0.1, 0.3]$ (ring-corrected), consistent with the empirical boundary. This is the dominant systematic bias; tightening it without losing tractability is the open problem.

Heuristic strategy space (`Firefighter` $\neq$ all-Rest). The empirical solver picks `Firefighter`s (`work_tendency`=0.9) as the "free-riders" in the asymmetric NE profile, not literal RESTers (`work_tendency`=0.2). Three `Firefighter`s at 90% WORK contribute non-zero expected extinguish probability—the effective k_{-i} in the lone-WORKER's deviation comparison is closer to $1+3\cdot 0.9 = 3.7$ than to the analytical k_{-i}=1. At κ=0.9, $\tilde{A}\kappa(1-\kappa)^{3.7} \approx 0.013 \ll c_{\text{gap}}$=1.0, so the lone WORKER should not exist as a NE under that interpretation. The empirical solver still reporting asymmetric_only at κ=0.9 suggests the per-agent ownership reward—not captured in the single-house mean-field reward—anchors each agent to its own house even when free-riding on others' fires. The asymmetric NE is therefore partly an artefact of per-agent ownership, not a pure Volunteer's-Dilemma equilibrium.

Per-agent ownership rewards (the W gap). The analytical A coefficient lumps r_{own}=50 and p_{own}=100 into the survival term, but the per-agent reward only fires on the agent's *own* house. The single-house reduction assumes the representative house is the agent's own house with probability 1; the ring-corrected reality is probability $1/H$=0.1. The other 9 houses contribute through the team term only, which scales A down by a factor of roughly $(p_{\text{own}}/H + P_{\text{team}}/H)/p_{\text{own}} \approx 0.11$ on the off-own-house terms. The ring-corrected A sits between A/H=181 and A=1812, depending on the agent's house-target distribution.

β-independence: exact by construction in bernoulli mode, not merely leading-order. Earlier drafts of this appendix treated β-independence as a leading-order approximation that could break via across-night contagion at large $\beta\rho$. Mechanistic inspection of the engine (repo issue #442) shows that under the canonical phase order in bernoulli extinguish mode the claim is *exact*: the burn-out phase (phase 4) transitions every still-BURNING house to RUINED before the spread phase (phase 5) runs, in every night, so spread never has a BURNING source, never fires, and draws zero RNG. The dynamics *and the RNG stream* are bit-identical across β; there is no across-night path for β to enter, and no ρ-sweep can activate it. Two consequences follow. First, the 39-cell grid's β axis carries no environmental information in this mode – a design lesson for future sweeps, which should either drop the axis or run a variant in which β is live (the contagion-active order that swaps phases 4 and 5, or the continuous-extinguish variant of §A.5). Second, observed cross-β differences in solver output at fixed (κ, c) – the splitting verdict row at (κ=0.9, c=0.5) and payoff deltas such as 80.9 vs. 72.0 at that cell and −614.4 vs. −648.0 at (κ=0.5, c=0.5) – are pure measurements of double-oracle solver nondeterminism, usable as a free noise probe.

Small-$q(k)$ linearisation. The survival integral of §B.2 approximates $(1-q(k))^{T_{\min}} \approx 1-T_{\min}\cdot q(k)$, a first-order Taylor step that is only accurate when $T_{\min}\cdot q(k) \ll 1$. At the parameter corner where the approximation is worst (κ small, so $q(k) \to \rho$=0.02 and $T_{\min}q \approx 0.24$), the linearisation overstates the expected loss by the neglected $O\big((T_{\min}q)^2\big)$ terms—i.e. precisely in the collapse band where the predicted threshold is already 3–10× too low. The bias is smaller than the ring-locality term but acts in the same region, so it is listed as a distinct source rather than folded into the survival coefficient.

Single-stage vs. multi-stage SPE. The reduction is the single-night stage game under stationary opponent policies, not the full multi-stage subgame-perfect equilibrium. Empirically, the DO solver evaluates policies over full $T_{\min}$=12 episodes, so per-cell payoffs reflect the integrated multi-stage outcomes. A symmetric all-Work profile that is a NE of the single-night stage game may fail to be a NE of the multi-stage game if a deviator can sustain a long-horizon advantage by switching to Rest at a specific night. The reduction folds multi-stage dynamics into A by integrating expected survival over $T_{\min}$ nights, but the exact multi-stage NE conditions involve subgame-perfect best-response calculations the single-night reduction cannot represent.

B.6 Implications and recommended follow-ups

Even with the quantitative thresholds wrong by 3–10×, the structural predictions the mean-field reduction makes—(a) κ-monotonicity of the verdict and (b) the existence of a no-pure-NE collapse regime at small κ—are empirically observed and would not be derivable without the closed-form derivation. (A third prediction, (c) β-independence under the canonical phase order, is correct but vacuously so in bernoulli extinguish mode, per §B.5; it is no longer counted as an empirical confirmation.) The body of the paper adopts inequalities (S), (A), (C) as the analytical anchor.

Three concrete follow-ups, in priority order: (1) complete the empirical grid across $c \in \{0.5, 1.0, 2.0\}$ and re-run the comparison table (now done at the 39-cell scale of §3); (2) re-run a β slice under a variant in which β is live—the contagion-active phase order or the continuous-extinguish mode of §A.5—so the reduction's β treatment can actually be falsified (in bernoulli mode it cannot be; repo issue #442); (3) derive a ring-Markov correction to A that replaces the single-house mean-field assumption with a chain-Markov approximation over the 10-house ring, closing the §B.5 gap without paying the full state-space cost.